

SAT Math Mastery · Week 4A**Quadratics II: Discriminant, Tangency & Vieta — Answer Key & Worked Solutions****Time**

120 minutes: 90 min solving + 30 min self-check and error log

Topicsnumber of solutions via the discriminant · tangent line to a parabola · sum and product of roots · nonlinear systems (line + parabola, line + circle) · parameter conditions · identities true for all x **Name / Date** _____**1 · SKILLS & IDEAS**

- S1.** Decide the number of real solutions of $ax^2 + bx + c = 0$ from $\Delta = b^2 - 4ac$ (also from a graph).
- S2.** Find a constant that makes an equation have exactly one, two, or no real solutions (strict vs. non-strict inequalities, integer counts).
- S3.** Use Vieta: sum $-\frac{b}{a}$ and product $\frac{c}{a}$ of the solutions without solving; build symmetric expressions such as $r^2 + s^2$.
- S4.** Substitute a line into a parabola or circle, then use Δ to count intersection points or force tangency.
- S5.** Treat “intersect at two points” problems with the sum and product of the x -coordinates (midpoint, distance).
- S6.** Handle hidden conditions: leading coefficient could be 0, “distinct” vs. “real”, sign conditions on roots.
- S7.** Match coefficients when two expressions are equal for all x (identities).
- S8.** Read the sign of a , b , c and Δ from the picture of a parabola.

2 · REQUIRED FACTS

- $\Delta > 0$: two distinct real solutions. $\Delta = 0$: exactly one real (repeated) solution. $\Delta < 0$: no real solutions.
- A line is tangent to a parabola (or circle) exactly when the substituted quadratic has $\Delta = 0$: one common point.
- For $ax^2 + bx + c = 0$ with solutions r and s : $r + s = -\frac{b}{a}$ and $rs = \frac{c}{a}$. These hold whether or not the solutions are nice.
- “No real solutions” $\iff$ the graph of $y = ax^2 + bx + c$ never touches the x -axis.
- If a and c have opposite signs, then $\Delta > 0$ (the parabola must cross the x -axis).
- If the leading coefficient contains a parameter, check the value that makes it 0: the equation stops being quadratic.

3 · REQUIRED FORMULAS

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad \Delta = b^2 - 4ac, \quad |r - s| = \frac{\sqrt{\Delta}}{|a|}$$

$$r + s = -\frac{b}{a}, \quad rs = \frac{c}{a}, \quad r^2 + s^2 = (r + s)^2 - 2rs, \quad r^3 + s^3 = (r + s)^3 - 3rs(r + s)$$

$$\text{vertex: } x = -\frac{b}{2a}, \quad (x - r)(x - s) = x^2 - (r + s)x + rs, \quad \text{one solution } \iff ax^2 + bx + c = a(x - h)^2$$

4 · MINI THEORY SHEET

Count solutions in three steps. (1) Move everything to one side: $ax^2 + bx + c = 0$. (2) Compute $\Delta = b^2 - 4ac$ (keep the constant k symbolic). (3) Turn the wording into an inequality on Δ .

Wording	Condition on Δ
exactly one (real) solution, tangent, touches	$\Delta = 0$
two distinct real solutions, crosses twice	$\Delta > 0$
no real solutions, never crosses / meets	$\Delta < 0$
at least one real solution	$\Delta \geq 0$
the <i>greatest integer</i> with no solution	largest integer satisfying the strict inequality

Line meets curve. Substitute the line $y = mx + k$ into the curve. The result is a quadratic in x whose Δ counts intersection points. Tangency $\iff \Delta = 0$.

Two intersection points. The x -coordinates are the two solutions, so their sum and product come straight from Vieta; the y -coordinates follow from the line.

Parameter in the leading coefficient. For $(k-1)x^2 + \dots = 0$, test $k = 1$ separately: the equation becomes linear and may have exactly one solution.

Identities. If two expressions are equal for every x , expand and match the coefficients of x^2 , x and the constants.

Reading a graph. Opens up $\Rightarrow a > 0$. Vertex right of the y -axis $\Rightarrow -\frac{b}{2a} > 0$, so a and b have opposite signs. y -intercept $= c$. Crosses the x -axis twice $\Rightarrow \Delta > 0$.

Speed shortcuts.

Situation	Shortcut
sum or product of solutions asked $r^2 + s^2$, $r^3 + s^3$, $\frac{r}{s} + \frac{s}{r}$	Vieta: no solving, no factoring; first rewrite as $ax^2 + bx + c = 0$ express with $r + s$ and rs
$x^2 + bx + c$ with exactly one solution	$c = \left(\frac{b}{2}\right)^2$ (perfect square)
$ r - s $ between two solutions	$\frac{\sqrt{\Delta}}{ a }$
line $y = mx + k$ and $y = x^2 + \dots$	set equal, one quadratic, then Δ
circle $x^2 + y^2 = R^2$ and line	line tangent $\iff$ distance from centre to line $= R$
number of integer k with $\Delta < 0$	$k^2 < N$ gives $-\sqrt{N} < k < \sqrt{N}$: count the integers carefully (strict)
$ax^2 + bx + c = 0$ with $ac < 0$	always two real solutions

5 · GUIDED EXAMPLES

Example 1 — Constant that gives exactly one solution

For what value of k does $3x^2 - 12x + k = 0$ have exactly one real solution?

$\Delta = (-12)^2 - 4(3)(k) = 144 - 12k$. Exactly one solution: $\Delta = 0 \Rightarrow k = 12$. (Check: $3x^2 - 12x + 12 = 3(x-2)^2$.)

Example 2 — Tangent line to a parabola

The line $y = 4x + k$ is tangent to the parabola $y = x^2 + 2x + 5$. Find k .

Set equal: $x^2 + 2x + 5 = 4x + k \Rightarrow x^2 - 2x + (5 - k) = 0$. Tangent $\Rightarrow \Delta = 0$: $4 - 4(5 - k) = 0 \Rightarrow k = 4$.
(The touching point: $x = 1$, $y = 8$.)

Example 3 — Vieta without solving

The solutions of $2x^2 - 9x + 3 = 0$ are r and s . Find $r^2 + s^2$.

$r + s = \frac{9}{2}$, $rs = \frac{3}{2}$. Then $r^2 + s^2 = (r + s)^2 - 2rs = \frac{81}{4} - 3 = \frac{69}{4}$.

6 · PRACTICE SET — 30 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

How many distinct real solutions does the equation $2x^2 - 5x + 4 = 0$ have?

(A) Infinitely many

(B) Zero ✓

(C) Exactly one

(D) Exactly two

Worked solution

1. $\Delta = b^2 - 4ac = (-5)^2 - 4(2)(4) = 25 - 32 = -7$.

2. $\Delta < 0$, so the equation has no real solutions.

Answer: (B)

2.

EASY · NO CALCULATOR NEEDED

What is the sum of the solutions of $x^2 - 9x + 14 = 0$?

Worked solution

1. By Vieta, the sum is $-\frac{b}{a} = -\frac{-9}{1} = 9$.

2. (Check: $(x - 2)(x - 7) = 0$, and $2 + 7 = 9$.)

Answer: 9

3.

EASY · NO CALCULATOR NEEDED

What is the product of the solutions of $3x^2 - 7x - 12 = 0$?

(A) $\frac{7}{3}$ (B) $-\frac{7}{3}$ **(C) -4 ✓**

(D) 4

Worked solution

1. By Vieta, the product is $\frac{c}{a} = \frac{-12}{3} = -4$.

2. The discriminant is $49 + 144 > 0$, so both solutions are real.

Answer: (C)

4.

EASY · NO CALCULATOR NEEDED

The equation $x^2 + 6x + c = 0$ has exactly one real solution, where c is a constant. What is the value of c ?

Worked solution

1. Exactly one solution means $\Delta = 0$: $6^2 - 4(1)(c) = 0$.

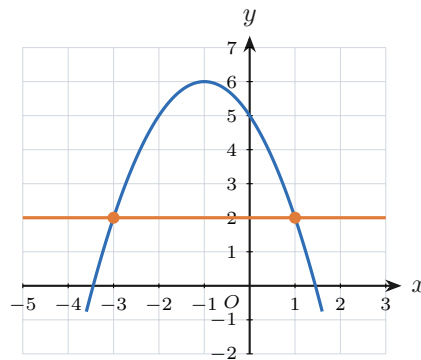
2. $36 - 4c = 0 \Rightarrow c = 9$.

Answer: 9

5.

EASY · DESMOS

The graph of a system of two equations in the xy -plane is shown. How many solutions does the system have?



(A) 2 ✓

(B) 3

(C) 0

(D) 1

Worked solution

1. Solutions of a system are the points where the two graphs meet.
2. The line $y = 2$ crosses the parabola at two points, $(-3, 2)$ and $(1, 2)$.
3. The system has 2 solutions.

Answer: (A)

6.

MEDIUM · NO CALCULATOR NEEDED

For which of the following values of k does the equation $2x^2 + kx + 8 = 0$ have no real solutions?

(A) -9 (B) -8 (C) 8

(D) 5 ✓

Worked solution

1. $\Delta = k^2 - 4(2)(8) = k^2 - 64$.
2. No real solutions: $k^2 - 64 < 0 \Rightarrow -8 < k < 8$.
3. Only 5 lies strictly between -8 and 8 . ($k = \pm 8$ gives exactly one solution; $k = -9$ gives two.)

Answer: (D)

7.

MEDIUM · NO CALCULATOR NEEDED

The equation $2x^2 + 8x + k = 0$ has no real solutions, where k is an integer. What is the least possible value of k ?

Worked solution

1. $\Delta = 8^2 - 4(2)(k) = 64 - 8k$.
2. No real solutions: $64 - 8k < 0 \Rightarrow k > 8$.
3. $k = 8$ gives $\Delta = 0$ (one solution), so it is excluded. The least integer is 9.

Answer: 9

8.

MEDIUM · NO CALCULATOR NEEDED

The equation $3x^2 - 5 = 11x + x^2$ has two real solutions. What is the sum of the solutions?

(A) $-\frac{11}{2}$

(B) $-\frac{5}{2}$

(C) $\frac{11}{2}$ ✓

(D) 11

Worked solution

1. Move all terms to one side: $3x^2 - x^2 - 11x - 5 = 0 \Rightarrow 2x^2 - 11x - 5 = 0$.

2. Sum = $-\frac{b}{a} = -\frac{-11}{2} = \frac{11}{2}$.

Answer: (C)

9.

MEDIUM · NO CALCULATOR NEEDED

In the xy -plane, the line $y = 2x + k$ is tangent to the parabola $y = x^2 + 4x + 7$, where k is a constant. What is the value of k ?

(A) 3

(B) 6 ✓

(C) 7

(D) -6

Worked solution

1. Set the expressions equal: $x^2 + 4x + 7 = 2x + k \Rightarrow x^2 + 2x + (7 - k) = 0$.

2. Tangent means exactly one common point: $\Delta = 4 - 4(7 - k) = 0$.

3. $4k - 24 = 0 \Rightarrow k = 6$.

Answer: (B)

10.

MEDIUM · DESMOS

In the xy -plane, the line $y = 6$ intersects the parabola $y = x^2 - 6x + 11$ at two points. What is the distance between the two points?

(A) 5

(B) 16

(C) 2

(D) 4 ✓

Worked solution

1. Set $x^2 - 6x + 11 = 6 \Rightarrow x^2 - 6x + 5 = 0 \Rightarrow (x - 1)(x - 5) = 0$.

2. The points are (1, 6) and (5, 6); they lie on a horizontal line.

3. Distance = $5 - 1 = 4$.

Answer: (D)

11.

MEDIUM · DESMOS

$$y = x^2 - 5$$

$$y = 2x + 3$$

The solutions to the given system of equations are (x_1, y_1) and (x_2, y_2) . What is the value of $y_1 + y_2$?

(A) 10 ✓

(B) 12

(C) 2

(D) 8

Worked solution

1. Substitute: $x^2 - 5 = 2x + 3 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x - 4)(x + 2) = 0$.

2. $x = 4 \Rightarrow y = 2(4) + 3 = 11$; $x = -2 \Rightarrow y = 2(-2) + 3 = -1$.

3. $y_1 + y_2 = 11 + (-1) = 10$. (Note $x_1 + x_2 = 2$ is a different quantity.)

Answer: (A)

12.

MEDIUM · NO CALCULATOR NEEDED

The solutions of $x^2 - 10x + c = 0$ are r and s , where $r > s$ and $r - s = 4$. What is the value of c ?

Worked solution

1. Vieta: $r + s = 10$. Given: $r - s = 4$.

2. Add: $2r = 14 \Rightarrow r = 7$, so $s = 3$.

3. $c = rs = 7 \cdot 3 = 21$.

Answer: 21

13.

MEDIUM · NO CALCULATOR NEEDED

A rectangular garden has a perimeter of 36 meters and an area of 80 square meters. What is the length, in meters, of the longer side of the garden?

Worked solution

1. Let one side be w ; the other is $18 - w$ (half the perimeter).

2. $w(18 - w) = 80 \Rightarrow w^2 - 18w + 80 = 0 \Rightarrow (w - 8)(w - 10) = 0$.

3. The sides are 8 and 10; the longer side is 10 meters.

Answer: 10

14.

HARD · NO CALCULATOR NEEDED

In the equation $5x^2 + (k - 3)x + 5 = 0$, k is a constant. For which of the following values of k does the equation have exactly one real solution?

(A) -10

(B) -7 ✓

(C) 7

(D) 10

Worked solution

1. $\Delta = (k - 3)^2 - 4(5)(5) = (k - 3)^2 - 100$.

2. $\Delta = 0 \Rightarrow k - 3 = \pm 10 \Rightarrow k = 13$ or $k = -7$.

3. Only -7 appears among the choices. ($k = \pm 10$ comes from forgetting the -3; $k = 7$ gives $(4)^2 - 100 \neq 0$.)

Answer: (B)

15.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the line $y = mx - 5$, where $m > 0$, is tangent to the parabola $y = 2x^2 + 3$. What is the value of m ?

Worked solution

1. Set equal: $2x^2 + 3 = mx - 5 \Rightarrow 2x^2 - mx + 8 = 0$.
2. Tangent: $\Delta = m^2 - 4(2)(8) = 0 \Rightarrow m^2 = 64 \Rightarrow m = \pm 8$.
3. Since $m > 0$, $m = 8$.

Answer: 8

16.

HARD · NO CALCULATOR NEEDED

The solutions of $x^2 - 7x + 5 = 0$ are r and s . What is the value of $r^2 + s^2$?

(A) 29

(B) 59

(C) 39 ✓

(D) 49

Worked solution

1. Vieta: $r + s = 7$ and $rs = 5$.
2. $r^2 + s^2 = (r + s)^2 - 2rs = 49 - 10 = 39$.

Answer: (C)

17.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the line $y = x + k$, where $k > 0$, is tangent to the circle $x^2 + y^2 = 50$. What is the value of k ?

Worked solution

1. Substitute: $x^2 + (x + k)^2 = 50 \Rightarrow 2x^2 + 2kx + (k^2 - 50) = 0$.
2. Tangent: $\Delta = (2k)^2 - 4(2)(k^2 - 50) = 400 - 4k^2 = 0 \Rightarrow k^2 = 100$.
3. Since $k > 0$, $k = 10$. (Check: distance from the origin to $x - y + 10 = 0$ is $\frac{10}{\sqrt{2}} = 5\sqrt{2} = \sqrt{50}$.)

Answer: 10

18.

HARD · NO CALCULATOR NEEDED

For how many integer values of k does the equation $3x^2 + kx + 12 = 0$ have no real solutions?

(A) 23 ✓

(B) 24

(C) 11

(D) 22

Worked solution

1. $\Delta = k^2 - 4(3)(12) = k^2 - 144 < 0 \Rightarrow -12 < k < 12$.
2. The integers strictly between -12 and 12 are $-11, -10, \dots, 11$.
3. That is $11 + 1 + 11 = 23$ values. (The endpoints ± 12 give $\Delta = 0$ and are excluded.)

Answer: (A)

19.

HARD · NO CALCULATOR NEEDED

The equation $x^2 - 9x + k = 0$ has two real solutions, and one solution is twice the other. What is the value of k ?

Worked solution

1. Let the solutions be r and $2r$. Then $r + 2r = 9 \Rightarrow r = 3$.
2. The solutions are 3 and 6.
3. $k = 3 \cdot 6 = 18$.

Answer: 18

20.

HARD · NO CALCULATOR NEEDED

In the equation $ax^2 + bx + c = 0$, a , b , and c are constants with $a > 0$ and $c < 0$. Which of the following must be true?

- I. The equation has two distinct real solutions.
- II. The two solutions are both positive.
- III. The product of the two solutions is negative.

(A) I, II, and III

(B) I only

(C) III only

(D) I and III only ✓

Worked solution

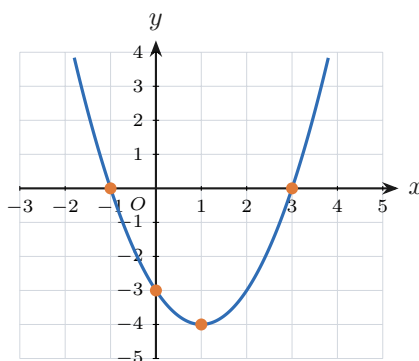
- $ac < 0$, so $\Delta = b^2 - 4ac = b^2 + (\text{positive}) > 0$: statement I is true.
- Product of solutions $= \frac{c}{a} < 0$: statement III is true.
- A negative product means the solutions have opposite signs, so II is false.
- Only I and III must be true.

Answer: (D)

21.

HARD · DESMOS

The graph of $y = f(x)$ is shown in the xy -plane. For which of the following values of k does the equation $f(x) + k = 0$ have no real solutions?

(A) -4 (B) 4 (C) 5 ✓(D) -5 **Worked solution**

- The vertex $(1, -4)$ shows the minimum value of f is -4 , so $f(x)$ takes every value ≥ -4 and no value below -4 .
- $f(x) + k = 0 \iff f(x) = -k$. No real solutions requires $-k < -4$, i.e. $k > 4$.
- Check: $k = 5$ gives $f(x) = -5 < -4$ (none); $k = 4$ gives $f(x) = -4$ (one solution); $k = -4$ and $k = -5$ give $f(x) = 4$ and 5 (two solutions each).

Answer: (C)

22.

HARD · CALCULATOR

A ball is thrown upward from a platform. Its height, in feet, t seconds after it is thrown is $h(t) = -16t^2 + 64t + 6$. The ball is at a height of exactly H feet at exactly one time during its flight, where $H > 6$. What is the value of H ?

Worked solution

1. Height H at one time only: $-16t^2 + 64t + 6 = H \Rightarrow 16t^2 - 64t + (H - 6) = 0$ has $\Delta = 0$.
2. $\Delta = 64^2 - 4(16)(H - 6) = 4096 - 64(H - 6) = 0 \Rightarrow H - 6 = 64$.
3. $H = 70$. (This is the maximum height: vertex at $t = 2$, $h(2) = -64 + 128 + 6 = 70$.)

Answer: 70

23.

HARD · NO CALCULATOR NEEDED

The equation $(ax + 3)(2x - b) = 6x^2 + cx - 12$ is true for all values of x , where a , b , and c are constants. What is the value of c ?

Worked solution

1. Expand the left side: $2ax^2 - abx + 6x - 3b = 2ax^2 + (6 - ab)x - 3b$.
2. Match x^2 : $2a = 6 \Rightarrow a = 3$. Match constants: $-3b = -12 \Rightarrow b = 4$.
3. Match x : $c = 6 - ab = 6 - 12 = -6$.

Answer: -6

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a constant k , the equation $(k - 1)x^2 - 6x + (k - 1) = 0$ has exactly one real solution. What is the sum of all possible values of k ?

Worked solution

1. If $k \neq 1$ the equation is quadratic with $\Delta = 36 - 4(k - 1)^2$. Exactly one solution: $(k - 1)^2 = 9 \Rightarrow k - 1 = \pm 3 \Rightarrow k = 4$ or $k = -2$.
2. If $k = 1$ the equation becomes $-6x = 0$, which is linear and has exactly one solution, $x = 0$. So $k = 1$ also works.
3. Sum of all possible values: $4 + (-2) + 1 = 3$.

Answer: 3

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

The solutions of $x^2 - 6x + 4 = 0$ are r and s . What is the value of $r^3 + s^3$?

(A) 108

(B) 144 ✓

(C) 216

(D) 72

Worked solution

1. Vieta: $r + s = 6$ and $rs = 4$.
2. $r^3 + s^3 = (r + s)^3 - 3rs(r + s)$.
3. $= 6^3 - 3(4)(6) = 216 - 72 = 144$.

Answer: (B)

26.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $y = x + 2$ intersects the circle $x^2 + y^2 = 20$ at points P and Q . What is the value of $(PQ)^2$?

Worked solution

1. Substitute: $x^2 + (x + 2)^2 = 20 \Rightarrow 2x^2 + 4x - 16 = 0 \Rightarrow x^2 + 2x - 8 = 0 \Rightarrow (x + 4)(x - 2) = 0$.
2. $P = (2, 4)$ and $Q = (-4, -2)$.
3. $(PQ)^2 = (2 - (-4))^2 + (4 - (-2))^2 = 36 + 36 = 72$.

Answer: 72

27.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $y = 2x - 3$ is tangent to the parabola $y = x^2 - 6x + c$, where c is a constant. The point of tangency is (p, q) . What is the value of $p + q$?

(A) 5

(B) 13

(C) 4

(D) 9 ✓**Worked solution**

1. Set equal: $x^2 - 6x + c = 2x - 3 \Rightarrow x^2 - 8x + (c + 3) = 0$.

2. Tangent: $\Delta = 64 - 4(c + 3) = 0 \Rightarrow c = 13$. The double solution is $x = \frac{8}{2} = 4$, so $p = 4$.

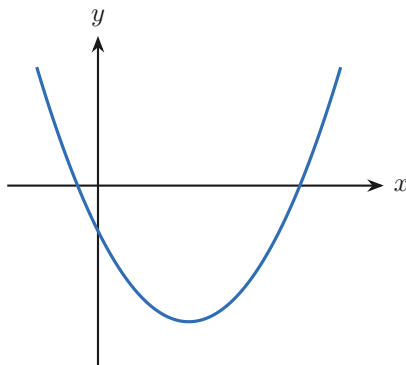
3. $q = 2(4) - 3 = 5$, so $p + q = 9$. (Check on the parabola: $16 - 24 + 13 = 5$.)

Answer: (D)

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

The graph of $y = ax^2 + bx + c$ is shown in the xy -plane, where a , b , and c are constants. Which of the following must be true?

I. $b^2 - 4ac > 0$ II. $ab < 0$ III. $c > 0$ **(A) I and II only ✓**

(B) II and III only

(C) I, II, and III

(D) I only

Worked solution

1. The parabola opens upward, so $a > 0$.

2. It crosses the x -axis twice, so $\Delta = b^2 - 4ac > 0$: I is true.

3. The vertex lies to the right of the y -axis: $-\frac{b}{2a} > 0$ with $a > 0$ gives $b < 0$, so $ab < 0$: II is true.

4. The graph crosses the y -axis below the origin, so $c < 0$: III is false. Only I and II are true.

Answer: (A)

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

What is the least integer k for which the equation $x^2 - kx + (k + 3) = 0$ has two distinct positive real solutions?

(A) 6

(B) 7 ✓

(C) 8

(D) 3

Worked solution

1. Two distinct real solutions: $\Delta = k^2 - 4(k + 3) = k^2 - 4k - 12 = (k - 6)(k + 2) > 0 \Rightarrow k > 6$ or $k < -2$.
2. Both solutions positive: sum $= k > 0$ and product $= k + 3 > 0$. Together with $\Delta > 0$ this forces $k > 6$.
3. The least integer is $k = 7$. ($k = 6$ gives $\Delta = 0$, a single repeated solution $x = 3$, so it fails the word “distinct”.)

Answer: (B)

30.

SAT CHALLENGE · DESMOS

In the xy -plane, the line $y = 2x + k$ intersects the parabola $y = x^2 - 4x + 1$ at two points whose x -coordinates differ by 8. What is the value of k ?

Worked solution

1. Set equal: $x^2 - 4x + 1 = 2x + k \Rightarrow x^2 - 6x + (1 - k) = 0$.
2. The x -coordinates x_1, x_2 satisfy $x_1 + x_2 = 6$ and $x_1 x_2 = 1 - k$. Given $|x_1 - x_2| = 8$: $(x_1 - x_2)^2 = (x_1 + x_2)^2 - 4x_1 x_2$.
3. $64 = 36 - 4(1 - k) \Rightarrow 4(1 - k) = -28 \Rightarrow 1 - k = -7 \Rightarrow k = 8$. (Check: $x^2 - 6x - 7 = 0$ gives $x = -1, 7$.)

Answer: 8